



DIBBs eCR Viewer

eCR Viewer FAQ

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**U.S. DEPARTMENT OF
HEALTH AND HUMAN SERVICES**
CENTERS FOR DISEASE
CONTROL AND PREVENTION



Version History

Version	Author	Reviewed By	Revision Date	Revision Notes
1.0	Elizabeth Chinelo Ikejimba	Kristie Clarke	09/17/2025	Revised to contain details for all 3 Viewer options (e.g. NBS-integrated, Standalone eCR Library, and dual boot mode)
1.1	Matthew Goldberg	Kristie Clarke	12/09/2025	Revised based on Kristie's feedback
1.2	Matthew Goldberg	Michael Wodajo	04/24/2026	Revised to update based on current process



Table of Contents

1	eCR Viewer Overview.....	4
2	Summarizes relevant information for the reportable condition(s)	5
3	Types of integrations	8
4	Authorization in the eCR Viewer	10
5	Technical overview	13
6	How to get help.....	13
7	Glossary.....	14

To contact the eCR Viewer team, email dibbs@cdc.gov

1 eCR Viewer Overview

Note: For definitions of any unfamiliar terms or acronyms (e.g., eCR), refer to the Glossary at the end of this document.

1.1 What is the eCR Viewer?

The eCR Viewer is a tool that provides a human-readable view of eCRs, making it easier for public health staff to find relevant information in eCR documents. It combines both the electronic initial case report (eICR) and Reportability Response (RR) into a single view and highlights relevant lab or clinical information for conditions present in the eCR. It is designed for case investigators, eCR coordinators, and others who review eCRs.

There are three eCR Viewer options:

- **Integrated eCR Viewer** - The eCR Viewer built directly into NBS (National Electronic Disease Surveillance System Base System) or EpiTrax, providing a standardized way to view an individual eCR within those systems
- **Standalone eCR Library** - A separate tool jurisdictions can use to manage and view eCR documents outside their surveillance system; it uses the same eCR Viewer as the integrated version for individual views of an eCR
- **Dual-boot mode** - Allows jurisdictions to access the eCR Viewer from two different entry points: either through NBS/EpiTrax (integrated) or through the Standalone eCR Library

Note: We will use the overarching term “eCR Viewer” to refer to all three versions of the product. When content is specific to a particular version, the full bolded term above will be used for clarity.

1.2 When does a jurisdiction need to choose an eCR Viewer option?

Jurisdictions select their eCR Viewer option during the Technical Scoping phase, which is the point when the technical requirements, staffing, and system capabilities are assessed. The decision can be changed later, but doing so requires an additional Technical Scoping session with the DIBBs team to confirm the jurisdiction still meets the personnel and technical requirements for the new option.

Please see phases of activity below:



1.3 What staff are required to deploy and operate the eCR Viewer?

Setting up the eCR Viewer requires a blend of programmatic oversight and technical execution. Depending on your jurisdiction's size, these five functions may be handled by five different people or consolidated into a single small team.

- **eCR Coordinator:** Manages internal eCR data flows, identifies subject matter experts from different program areas for testing, and ensures the Viewer meets the needs of the end-users.
- **Testing Lead:** Conducts system health checks, reviews technical acceptance documents, and validates data imports to ensure the system is reliable before the staff starts using it.
- **Systems Admin:** Configures account permissions, enables API access for automated tools (like Terraform), and works alongside the Cloud Implementation Lead to troubleshoot high-level networking or security blockers.
- **Cloud Implementation Lead:** Deploys the actual cloud resources, manages infrastructure code, and handles the technical "heavy lifting" of the deployment and environment debugging.
- **Integration Lead:** Connects the eCR Viewer to your existing data systems (e.g., Rhapsody). They ensure that the DIBBs pipeline is correctly talking to your jurisdiction's internal environment.

1.4 What challenges was the eCR Viewer designed to solve?

Users have communicated to CDC that they currently need to search through lengthy documents for information relevant to a case to find the information they need in an eICR and RR.

The eCR Viewer combines eICR and RR data together, enabling public health staff to see all of the reportable conditions for the eCR, why the eCR was triggered, and which jurisdictions the eCR was sent to, none of which are possible to view in the eICR data alone.

The eCR Viewer also pulls out data fields from the eICR XML, allowing public health staff to see more data than what is currently available by default in the HTML or surveillance system views.

1.5 What are the benefits of the eCR Viewer?

The eCR Viewer was specifically designed and co-created with public health departments and their epidemiologists. The tool offers several features that make eCR data more usable for jurisdictions.

2 Summarizes relevant information for the reportable condition(s)

Senator Mon Mothma 01/01/1975

eCR Summary

eCR Document

Patient Info

Demographics

Pregnancy Info

Encounter Info

Encounter Details

Facility Details

Provider Details

Clinical Info

Clinical Notes

Symptoms and Problems

Lab Info

Lab Results from Castle Lab

eCR Metadata

RR Details

eCR Details

eCR Custodian Details

Unavailable Info

eCR Summary

Provides key info upfront to help you understand the eCR at a glance

Patient Summary

Patient Name

Senator Mon Mothma

DOB

01/01/1975

Sex

Female

Race

White

Ethnicity

Not Hispanic or Latino

Patient Address

500 Republica, Suite 400
Senate District, Galactic City GC-500
Coruscant
County: Administrative Core

Patient Contact

Mobile: 555-555-9876
mon.mothma@senate.gr.example.com

Encounter Summary

Encounter Date/Time

Start: 05/13/2020 9:01 AM
End: 05/13/2020 10:47 AM

Encounter Type

Ambulatory

Encounter Diagnosis

Viral hepatitis type C (disorder)

Encounter Reason

Swollen legs

Facility Name

Grand Republic Medical Facility Clinic

Facility Contact

Work: 1-999-999-9999

Condition Summary

2 CONDITIONS FOUND

Viral hepatitis type C (disorder)

+

Anthrax (disorder)

RCKMS Rule Summary

Anthrax vaccine administered

Immunizations Relevant to Reportable Condition

Name	Administration Dates	Dose Number	Manufacturer	Lot Number
Anthrax	05/13/2020	1	Pfizer	LOT9911Z

Lab Results Relevant to Reportable Condition

No matching lab results found in this eCR

The eCR summary surfaces information to the top of the eCR document and also houses a condition summary.

Why is this important?

The summary provides critical information about an eCR at a glance. Any staff member can quickly find the information they need.

What does it include?

- **Patient summary** – Displays patient demographics and contact information
- **Encounter summary** – Displays encounter information like date/time and facility
- **Condition summary** - Highlights relevant information to reportable conditions that may have triggered the eCR; identifies the Reportable Conditions Knowledge Management System (RCKMS) rule summary as well as immunization, clinical, and lab information relevant to the condition

Page 6 of 14

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2.1.1 Keeps data formats consistent regardless of which EHR system generated the eCR

eCR Document
Displays entire eCR and RR documents to help you dig further into eCR data

Patient Info

Demographics

Patient Name: Master Minch Yoda
DOB: 01/01/1025
Age at Encounter: 900 years
Vital Status: Alive
Sex: Male
Ethnicity: Not Hispanic or Latino
Preferred Language: English
Good
Expressed opinion

Patient Address: Jedi Temple, Temple State
Temple District, Galactic City DC 101
Coruscant
County Administrative Core

Recent County: Administrative Core
Contact: 555-555-5555
master.yoda@jediorder.gov/example.com
Patient IDs: MFIN-00884411
000-55-54466

Social History

Travel History

Location	Date	Purpose	Details
Naali	Start: 01/15/2025 End: 02/05/2025	No data	No data

Each eCR comes in the same format, which includes the following sections: eCR Summary, Patient Info, Encounter Info, Clinical Info, Lab Info, eCR Metadata, and Unavailable Info.

Why is this important?

The formatting of an eCR is dependent on the provider or electronic health record (EHR), and the information order may vary. With the eCR Viewer, all the eCR information comes in a standardized layout regardless of the provider or the EHR system that generated the eCR.

What does it include?

- **Side menu** - Enables users to jump from one section of an eCR to another
- **Clear sections** - Arranges information into clear and easily navigable sections, including Patient Info, Encounter Info, Lab Info, eCR Metadata, and Unavailable Info (i.e., indicates when expected information is missing from an eCR; any missing data elements are listed in this section)
- **Structured fields** - Standardizes the format of complicated fields; for example, lab data is consolidated into a table and arranged in chronological order by receipt date

2.1.2 Combines information from both the eICR and the RR in 'eCR Metadata' section

Why is this important?

Epidemiologists reviewing the eCR can quickly understand why it was sent to their jurisdiction. This information isn't visible in the eICR data alone.

What does it include?

- **eCR Metadata** - Provides RR details, Electronic Reporting and Surveillance Distribution (eRSD) service warnings, eICR details, eICR Author details for the Practitioner, and eICR custodian details

3 Types of integrations

As mentioned, there are three eCR Viewer options:



- Integrated eCR Viewer
- Standalone eCR Library
- Dual-boot mode

3.1 What is the Integrated eCR Viewer?

The Integrated eCR Viewer is a direct integration of the core Viewer page within the user interface of an existing jurisdictional surveillance system. This is currently available for NBS (6.0.16.2). EpiTrax integration will be available in 2026.

The screenshot displays the NBS eCR Viewer interface. At the top, there's a navigation bar with links like Home, Data Entry, Merge Patients, Open Investigations, Reports, and System Management. A user profile section shows 'User: demo user' and 'NBS'. Below this, a 'View Document' section highlights the 'View eCR Document' button, which is circled in red. The main content area shows patient details for 'Sen. Mon Mothma | Female | 01/01/1975' with Patient ID: 84455. It includes fields for Jurisdiction, Program Area, Created/Last Updated dates, and By fields. A 'Public Health Case Report - Data From Legacy System To CDA' section follows, with links to various information tabs. The 'SENDING SYSTEM INFORMATION' section shows details about the message type, facility, creation date, and status. The 'PATIENT INFORMATION' section lists patient name, date of birth, race, sex, and contact info. The 'SOCIAL HISTORY INFORMATION' section includes birth sex and deceased status. The 'CLINICAL INFORMATION' section shows investigation ID, pregnancy status, condition, and physician details.

When viewing a patient record in NBS, the “View eCR Document” button (circled for emphasis) takes users to the individual patient eCR using the eCR Viewer.

The screenshot shows the eCR Viewer interface for 'Senator Mon Mothma | 01/01/1975'. The 'eCR Summary' section provides key information at a glance. It includes a 'Patient Summary' with fields for Patient Name, DOB, Sex, Race, Ethnicity, Patient Address, and Patient Contact. Below this is an 'Encounter Summary' with fields for Encounter Date/Time, Encounter Type, Facility Name, and Facility Contact. A 'Condition Summary' section is also visible at the bottom. The interface is clean and organized, with a sidebar on the left for navigation.

This is an example of the individual patient eCR using the eCR Viewer.

3.2 What is the Standalone eCR Library?

The Standalone eCR Library is a tool for users to view eCR documents outside of their jurisdiction's surveillance system. Columns include received date, encounter date, reportable condition, and RCKMS rule summary. This version creates a database of eCR / RR data in a unified, human-readable form. It uses the same eCR Viewer as the integrated version for individual views of an eCR. Clicking the name of the patient in the eCR Library will take the user to view the individual patient record.

eCR Viewer				
eCR library User management Program management				
eCR Library Search by patient 				
FILTERS: Last 24 hours Reportable Condition Reset				
Patient	Received Date	Encounter Date	Reportable Condition	RCKMS Rule Summary
Mon Mothma DOB: 01/01/1975	05/13/2020 9:01 AM	05/13/2020 9:01 AM	Viral hepatitis type C (disorder)	Detection of Hepatitis C virus antibody in a clinical specimen by any method
			Anthrax (disorder)	Anthrax (as a diagnosis or active problem)
				Anthrax vaccine administered
Dash Rendar DOB: 10/01/1955	05/13/2020 8:45 AM	05/13/2020 8:45 AM	Overdose of opiate (disorder)	Detection of opioid substance or metabolite in a clinical specimen by any method
Minch Yoda DOB: 01/01/1125	05/13/2020 8:00 AM	05/13/2020 8:00 AM	Malignant neoplastic disease (disorder)	Malignant or in situ neoplasm, or a benign or borderline primary intracranial or central nervous system tumor (as a diagnosis or active problem)
Han Solo DOB: 12/20/1990	05/13/2020 7:04 AM	05/13/2020 7:04 AM	Salmonella infection (disorder)	Detection of Salmonella species and serovars nucleic acid (except S. typhi and S. paratyphi nucleic acid) in a clinical specimen by any method
Anakin Skywalker DOB: 01/10/1961	05/13/2020 6:16 AM	05/13/2020 6:16 AM	Toxic effect of carbon monoxide (disorder)	Detection of carboxyhemoglobin (COHb) level $\geq 2.5\%$ by any method in a blood specimen
Jabba the Hutt DOB: 01/01/1974	05/13/2020 3:04 AM	05/13/2020 3:04 AM	Influenza caused by Influenza A virus subtype H5N1 (disorder)	Cough
				Fever (Temperature $> 37.8^{\circ}\text{C}$ [100°F])
Jar Jar Binks DOB: 10/10/1981	05/13/2020 12:21 AM	05/13/2020 12:21 AM	Malignant neoplastic disease (disorder)	Malignant or in situ neoplasm, or a benign or borderline primary intracranial or central nervous system tumor (as a diagnosis or active problem)
Showing 1-25 of 757 eCRs 1 2 3 4 ... 38 Next > eCRs per page 25 				

The eCR Library provides a table view of eCRs, including filters and search.

The eCR Library features include:

- Default initial grouping of eCRs for each patient by encounter date, from most to least recent
- Ability to search by patient
- Ability to search, filter, and sort eCRs by received date, encounter date or reportable condition
- Management of standard user permissions by program areas and/or reportable conditions by system administrators

3.3 What is Dual-boot mode?

“Dual-boot mode” allows jurisdictions to access the eCR Viewer from two different entry points: the Integrated eCR Viewer within NBS/Epitrax and the Standalone eCR Library where users can filter, sort, and search eCR documents separate from their surveillance system.

4 Authorization in the eCR Viewer

4.1 How does authentication work?

To access the eCR Viewer, all users will go through proper authentication. The authentication process differs depending on the eCR Viewer option utilized by the jurisdiction (Integrated eCR Viewer, Standalone eCR Library, or Dual-boot mode).

4.1.1 Authentication: Integrated eCR Viewer in NBS

NBS authentication will work as it already does today: NBS creates a JSON web token (JWT) from a public/private key pair shared between the eCR Viewer and NBS, which authenticates the request to the Viewer. If a user request comes via NBS, the eCR Viewer automatically assumes the request is valid.

Put another way, the eCR Viewer “piggybacks” on the Role-Based Access Control (RBAC) structures present in NBS and assumes that if a user has clicked the “View eICR Document” button in NBS, then the request must be valid. Critically, NBS users will not have to log in independently to the eCR Viewer if their request comes from NBS.

4.1.2 Authentication: Integrated eCR Viewer in Epitrax

Authentication in Epitrax will mirror authentication in NBS as closely as possible. In a similar vein, Epitrax users will not have to log in independently to the eCR Viewer if their request comes from Epitrax.

4.1.3 Authentication: Standalone eCR Library

Authentication to the eCR Library is supported with an integration either through [Keycloak](#) or [Active Directory \(AD\)](#). Using this method, users can log in with existing PHA credentials managed in AD/Keycloak, rather than the eCR Viewer managing credentials directly.

Users will need to add the eCR Viewer as an application to their authentication provider and set a number of environment variables on the eCR Viewer to allow it to connect. Find out more about these environment variables in the [eCR Viewer API Documentation](#) under “Authentication – Non-integrated.”

4.2 How are permissions managed within the Standalone eCR Library?

The Standalone eCR Library also allows for customization with user management and program management.

With regards to permissions, there are two main users:

- **Admins** - Admins manage the eCR Viewer and have access to the eCR Library, user management, and program management. They set and revoke user privileges based on program areas.
- **Standard users** - Standard users only access the eCR Library, which is populated with eCRs that are specific to their permissions. Permissions are not divided solely by individual diseases or disease groups; instead, they are controlled by program areas created and assigned by admins. A standard user can only see eCRs within the program areas they’re assigned—whether that

area contains a single condition (e.g., HIV/AIDS) or a broader group of conditions (e.g., Bloodborne or Respiratory Diseases).

4.2.1 User Management

eCR Viewer

eCR library
[User management](#)
Program management

User management

Create user

FILTERS:

All users

Program area 2

Email	User type	Program areas	Last logged in
example@philly.com	Standard	VPD	
test@example.com	Standard	Respiratory	
viewer_demo@dibbs.tools	Standard	All Conditions	05/13/2020 3:59 PM CST
viewer_demo_admin@dibbs.tools	Admin	All program areas	05/12/2020 3:41 PM CST

Showing 1-4 of 4 users

1

Users per page
10

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Admins manage eCR Library user access to records through the User management page.

User management enables Admins to create, modify, and delete user accounts as well as manage their access rights and permissions.

4.2.2 Program Management

eCR Viewer
 eCR library User management **Program management**

Program management

Create program area

Program area	↑	Number of conditions
All Conditions		213 conditions
Cancer		1 condition
Enteric		20 conditions
Respiratory		17 conditions
VPD		24 conditions

Showing 1-5 of 5 program areas

1

 Program areas per page 10

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Admins manage program areas and reportable conditions in the Program Management page.

Program management allows Admins to create program areas and assign specific reportable conditions to the program area. Admins can name the program area and utilize the customization to limit or expand user access to eCRs in the eCR Library. As an example, an epidemiologist assigned to the newborn screening program area would be able to see eCRs reported for birth defects, but not eCRs reported for COVID-19, carbon monoxide, etc.

At present, these permissions do not filter out any information in eCRs that may be outside the scope of a user's program area. If an eCR is dual-reported for TB and salmonella, for example, a user in the foodborne program area would still be able to see that the eCR was also reported for COVID-19.

These permissions only apply to the eCR Library version of the experience, so NBS users will be unaffected. As stated above, if a request comes directly from NBS, the eCR Viewer will fill it because it maps into the NBS permissions.

4.2.3 Audit Logging

The eCR Library creates an audit log table that logs the following actions:

- Processing an eCR to the Viewer
- Viewing an eCR
- Initial admin creation
- User sign in and sign out
- Creating, editing, or deleting users
- Creating, editing, or deleting program area

5 Technical overview

5.1 What basic technical requirements are needed for eCR Viewer?

To implement an eCR Viewer option, jurisdictions must verify the following technical prerequisites:

- **Surveillance System:** Document the specific System Name and Current Version
- **eCR Capabilities:** Confirm the ability to receive eCRs and author rules in RCKMS
- **Environment:** Identify the hosting provider (AWS, Azure, GCP, or On-premises).
- **Network Security:** Review internal constraints, including Firewall rules, VPC configurations, and connectivity requirements.
- **Data Transport:** Confirm an Integration Engine (e.g., Rhapsody, Mirth) is available to route messages.
- **Deployment Method:** Decide on deployment method (Terraform, scripts)

6 How to get help

6.1 How do we contact the DIBBs team?

If your jurisdiction is getting active support, email your point of contact on the DIBBs team. If not, email us directly at dibbs@cdc.gov for any other support needs.

6.2 What are some additional resources we can use to get started?

- [Setup Guide](#)
- [Database Documentation](#)
- [API Documentation](#)
- [Technical Acceptance Testing](#)
- [Audit Log Compliance](#)

7 Glossary

API: Application programming interface (API) is a set of functions and procedures that allow the creation of applications that access features and or data of another operating system, application, or other service.



eCR: electronic case reporting (eCR) is the automated, real-time exchange of case report information between electronic health records (EHRs) and public health agencies.

eICR: electronic initial case report (eICR) sent from a clinical care provider to determine if the condition code is reportable for its jurisdiction.

EHR: electronic health records (EHR) are the electronic version of a patient's medical history that providers maintain over time and may include all of the key administrative clinical data relevant to that patient's care.

eRSD: electronic reporting and surveillance distribution service (eRSD) is a service provided by APHL that looks at the EHR to determine if there's a trigger code present by using LOINC codes.

LOINC: Logical observation identifiers names and codes (LOINC) is a standardized coding system used for identifying health measurements, observations, and services. LOINC codes facilitate the exchange and aggregation of clinical laboratory data.

NBS: NEDSS Base System (NBS) is a CDC-developed integrated information system that helps STLT public health departments manage reportable disease data and send notifiable disease data to the CDC.

NEDSS: National Electronic Disease Surveillance System (NEDSS) is a set of architectural standards to support NNDSS.

RCKMS: Reportable Conditions Knowledge Management System (RCKMS) is an authoritative, real-time, digital portable to enhance disease surveillance by providing comprehensive information on public health reporting requirements to public health reporters such as clinicians and laboratories.

RR: Reportability response (RR) is a document, sent from a jurisdiction to a clinical care provider, that communicates information on the status of a reportable condition/

Schema: A schema is how a database categorizes recorded data.

STLT: State, tribal, local, and territory (STLT)

SQL: Structured query language (SQL) is a programming language for interacting with a database.